MAT 221 - CALCULUS I
Lesson 14 - Numerical Integration

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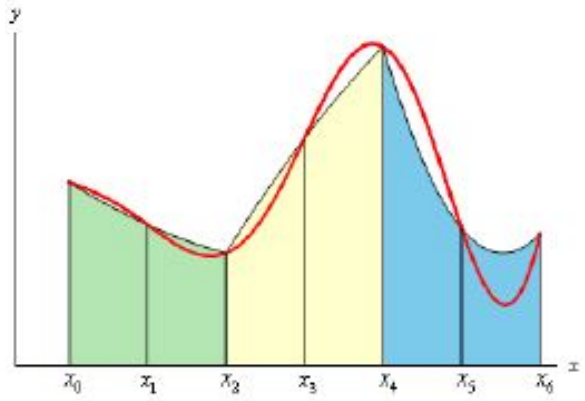
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Simpson's Rule

- In the Trapezoidal rule we approximated the curve with a straight line.
- For Simpson's rule we are going to approximate the function with a quadratic and we're going to require that the quadratic agree with three of the points from our subintervals.
- In this case we will again divide up the interval $[a, b]$ into n subintervals. However, unlike the previous method, we need to require that n be **even**.
- The width of each subinterval is

$$\Delta x = \frac{(b-a)}{n}.$$

- Below is a sketch of this using $n = 6$. Each of the approximations is colored differently so we can see how they actually work.



Notice that each approximation actually covers two of the subinterval. This is the reason for requiring n to be even.
It can be shown that the area under the approximation on the intervals $[x_{i-1}, x_i]$ and $[x_i, x_{i+1}]$ is

$$A_i = \frac{\Delta x}{3}(f(x_{i-1}) + 4f(x_i) + f(x_{i+1}))$$

If we use n subintervals the integral is then approximately,

$$\int_a^b f(x)\,dx \approx \frac{\Delta x}{3}(f(x_0) + 4f(x_1) + f(x_2)) + \frac{\Delta x}{3}(f(x_2) + 4f(x_3) + f(x_4)) + \ldots + \frac{\Delta x}{3}(f(x_{n-2}) + 4f(x_{n-1}) + f(x_n))$$

Theorem 1

This leads to the Simpsons rule to the definite integral given by;

$$\int_a^b f(x)\,dx \approx \frac{\Delta x}{3}[f(x_0 + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + \ldots + 2f(x_{n-2}) + f(x_n)].$$

Error Bound for Simpson's Rule

Suppose f is continuous on $[a, b]$ and suppose that f', f'', f''' and $f^{(4)}$ all exist on $[a, b]$, then the error bound for the Simpson's rule is given by

$$|\epsilon_n^S| \leqslant M\frac{(b-a)^5}{180n^4} \tag{1}$$

where $M = \max_{x\in[a,b]} |f^{(4)}(x)|$.

Example 1

(a) Estimate

$$\int_0^{\frac{\pi}{4}} \frac{1}{\cos x}\,dx$$

by using the Simpsons rule with 4 sub-intervals.

(b) What is the maximum error in using $n = 4$ to estimate

$$\int_0^1 \frac{1}{e^x}\,dx.$$

Solution

(a) $n = 4$, and $\Delta x = \frac{b-a}{n} = \frac{\frac{\pi}{4}-0}{4} = \frac{\pi}{16}$. Then

x_i	$f(x_i)$
0	1
$\frac{\pi}{16}$	1.0196
$\frac{\pi}{8}$	1.0824
$\frac{3\pi}{16}$	1.2027
$\frac{\pi}{4}$	1.4142

Hence

$$\int_0^{\frac{\pi}{4}} \frac{1}{\cos x} dx \approx \frac{1}{3}.\frac{\pi}{16}[1+4(1.0196)+2(1.0824)+4(1.2027)+1.4142]$$

that is,

$$\int_0^{\frac{\pi}{4}} \frac{1}{\cos x} dx \approx \frac{\pi}{48}(1+4.0784+2.1648+4.8108+1.4142) = .$$

(b) To calculate the maximum possible error of $\int_0^1 \frac{1}{e^x} dx$. we have, $f'(x) = -e^{-x}, f''(x) = e^{-x}, f'''(x) = -e^{-x}$ and $f^{(4)} = e^{-x}$. Therefore $M = e^0 = 1$.

$$|\epsilon_n^S| \leqslant 1.\frac{(1-0)^5}{180(4)^4} \approx 0.0000217. \qquad (2)$$

Example 2

Use the Simpson's rule to estimate

$$\int_0^2 e^{x^2} dx,$$

with a maximum error of 1.

Solution

(a) We need to find the step size that will produce a maximum error of 1 using Simpson's rule. We have $f'(x) = 2xe^{x^2}, f''(x) = (2 + 4x^2)e^{x^2}, f'''(x) = (12x + 8x^3)e^{x^2}$ and $f^{(4)} = (12 + 48x^2 + 16x^4)e^{x^2}$. Hence, M is at $x = 2$ in the interval $[0, 2]$, that is,$M = f^{(4)} = 460e^4 \approx 25115$. The error bound is

$$|\epsilon_n^S| \leqslant (25115) \times \frac{(2-0)^5}{180(n)^4} \approx \frac{4465}{n^4} < 1. \tag{3}$$

This means we need $n^4 > 4465$, i.e $n > 8.174$. The smallest even integer n that satisfies this condition is $n = 10$.

(b) n should be chosen large enough so that $|\epsilon_n^T| \leqslant 1$. We have $f(x) = e^{x^2}, f'(x) = 2xe^{x^2}$ and $f''(x) = (2 + 4x^2)e^{x^2}$. Therefore our M occurs at $x = 2$. That is, $M = f''(x = 2) = 18e^4 \approx 983$. Hence,

$$|\epsilon_n^T| \leqslant (983)\frac{(2-0)^3}{12(n)^2} \approx \frac{655}{n^2} \leqslant 1.$$

This gives $n \geqslant \sqrt{655}$. The smallest n that meets the requirement is $n = 26$.

Example 3

Use Simpson's rule to approximate the integral

$$\int_1^2 \ln(1 + x)\, dx$$

with a maximum error of 0.000001.